

~\Downloads\DAY_11.sql

```
1  -- Day 11: Infer schema and generate column descriptions from staged CSV data
2  -- Co-authored with CoCo
3  CREATE DATABASE IF NOT EXISTS DAY11_DB;
4
5  USE DATABASE DAY11_DB;
6
7  CREATE SCHEMA IF NOT EXISTS DAY11_SCHEMA;
8
9  USE SCHEMA DAY11_SCHEMA;
10
11 CREATE OR REPLACE FILE FORMAT CUSTOMER_FORMAT
12 TYPE = 'CSV'
13 FIELD_DELIMITER='|'
14 PARSE_HEADER=TRUE;
15
16 CREATE OR REPLACE STAGE CUST_STAGE
17 FILE_FORMAT=CUSTOMER_FORMAT;
18
19 SHOW STAGES;
20
21 list @CUST_STAGE;
22
23
24 SELECT
25 $1,
26 $2,
27 $3,
28 $4,
29 $5
30 FROM @CUST_STAGE/cust.csv;
31
32
33 SELECT *
34 FROM TABLE(
35 INFER_SCHEMA(
36     LOCATION=>'@CUST_STAGE',
37     FILES=>('cust.csv'),
38     FILE_FORMAT=>'CUSTOMER_FORMAT'
39 ));
40
41
42 -- Generate column description as text (for reference)
43 SELECT GENERATE_COLUMN_DESCRIPTION(
44     ARRAY_AGG(OBJECT_CONSTRUCT(*)), 'TABLE'
45 ) AS COLUMNS
46 FROM TABLE(
47 INFER_SCHEMA(
48     LOCATION=>'@CUST_STAGE',
```

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49 FILES=>('cust.csv'),
50 FILE_FORMAT=>'CUSTOMER_FORMAT'
51 ));
52
53 -- Create table using inferred schema template
54 CREATE OR REPLACE TABLE CUST_TABLE
55 USING TEMPLATE (
56     SELECT ARRAY_AGG(OBJECT_CONSTRUCT(*))
57     FROM TABLE(
58         INFER_SCHEMA(
59             LOCATION=>'@CUST_STAGE',
60             FILES=>('cust.csv'),
61             FILE_FORMAT=>'CUSTOMER_FORMAT'
62         )
63     )
64 );
65
66 -- Load data using MATCH_BY_COLUMN_NAME (required when PARSE_HEADER=TRUE)
67 COPY INTO CUST_TABLE
68 FROM @CUST_STAGE/cust.csv
69 FILE_FORMAT = 'CUSTOMER_FORMAT'
70 MATCH_BY_COLUMN_NAME = CASE_INSENSITIVE;
71
72
73 SELECT *
74 FROM CUST_TABLE;
75
76
77 SELECT
78 METADATA$FILENAME,
79 METADATA$FILE_ROW_NUMBER,
80 $1,
81
82 $2,
83
84 $3
85
86 FROM @CUST_STAGE/cust.csv;
```